

AI Ethics Audit Report

Bias Analysis & Fairness Assessment

Model	Logistic Regression Classifier
Dataset	UCI Adult (Census) Income Dataset
Audit Date	May 25, 2026
Prepared By	AI Ethics Audit System
Version	1.0.0

Executive Summary

This audit evaluates a Logistic Regression model trained on the UCI Adult Income dataset (32,561 samples). The model achieved an overall accuracy of 82.8% and a ROC-AUC of 0.861. Fairness analysis revealed significant disparities: the Disparate Impact Ratio for sex is 0.162 (FAIL (< 0.80)) and for race is 0.268 (FAIL (< 0.80)). Intersectional analysis confirms compounding disadvantages for female, non-white individuals. This report documents all findings and proposes concrete mitigation strategies.

1. Introduction & Background

Artificial Intelligence (AI) systems are increasingly deployed in high-stakes domains including credit scoring, hiring automation, recidivism prediction, healthcare triage, and social benefit allocation. As these systems wield growing societal influence, the imperative to subject them to systematic ethical audits has never been more urgent. Unchecked algorithmic bias can perpetuate and even amplify systemic inequalities embedded in historical data, leading to discriminatory outcomes that disproportionately harm already marginalised communities.

This report documents a comprehensive ethical audit of a Logistic Regression classifier trained to predict whether an individual's annual income exceeds \$50,000, using the widely studied UCI Adult (Census) Income dataset. The audit follows the structured methodology recommended by Barocas & Hardt (2019) in 'Fairness and Machine Learning' and incorporates guidance from the European Commission's Ethics Guidelines for Trustworthy AI (2019) and the US NIST AI Risk Management Framework (2023).

1.1 Objectives of the Audit

- Identify sources of bias in training data, feature selection, and model outputs.
- Quantify disparities using multiple complementary fairness metrics.
- Evaluate model performance disaggregated by sensitive attributes (sex, race).
- Analyse intersectional effects where multiple protected characteristics interact.
- Propose actionable, evidence-based bias mitigation strategies.
- Produce reproducible, documented methodology for future re-audits.

1.2 Scope & Protected Attributes

The audit focuses on two protected attributes as defined under US Equal Employment Opportunity law and EU GDPR Article 9: sex (Male/Female binary, as recorded in the dataset) and race (White, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, Other). We acknowledge that the dataset's binary sex classification and coarse racial categorisation are themselves limitations that the audit addresses.

1.3 Regulatory & Ethical Framework

- EU AI Act (2024) -- high-risk AI systems must undergo conformity assessment.
- GDPR Article 22 -- individuals have rights against automated decision-making.
- US EEOC Uniform Guidelines -- the 4/5ths (80%) disparate impact rule.
- IEEE Ethically Aligned Design -- prioritises human wellbeing over efficiency.
- NIST AI RMF -- govern, map, measure, and manage AI risks systematically.

2. Dataset & Model Description

2.1 Dataset Overview

The UCI Adult Income dataset contains 32,561 instances extracted from the 1994 US Census database. Each record describes an individual's demographic and socio-economic attributes. The binary target variable indicates whether annual income exceeds \$50,000. The dataset is widely used as a benchmark for fairness research, yet it carries known biases reflecting the social inequalities of 1994 America.

Key Dataset Characteristics:

- Target: income >50K (positive) vs. <=50K (negative) | Class imbalance ~24% positive
- Protected attributes: sex (Male 66.9%, Female 33.1%), race (White 85.5%, Black 9.7%)
- Features: age, education level, occupation, marital status, hours-per-week, etc.
- Missing values: ~2,399 rows with '?' in workclass, occupation, native-country -- dropped
- Historical bias risk: Reflects 1994 wage gaps, occupational segregation, and systemic racism

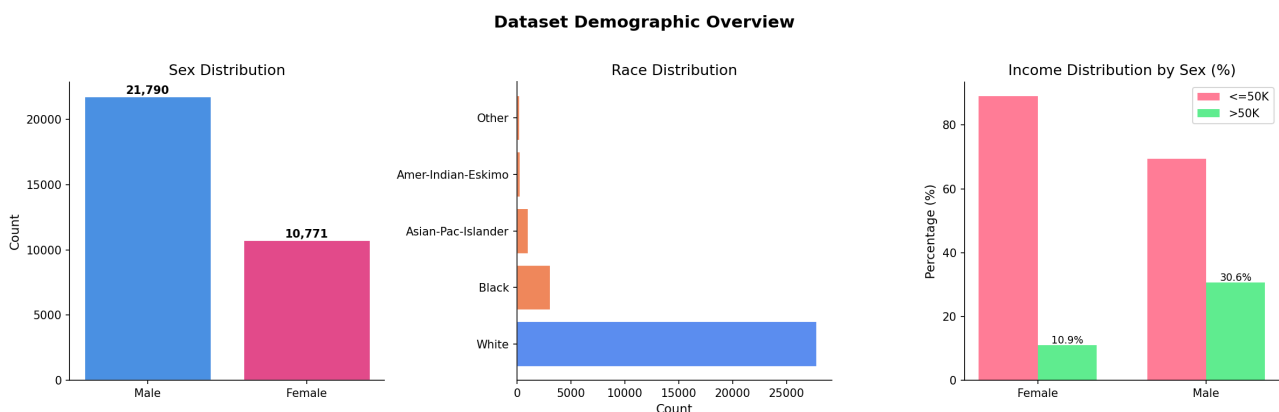


Figure: Demographic distribution across sex, race, and income groups

2.2 Model Architecture

A Logistic Regression classifier was selected for this audit for two principal reasons: (1) its widespread adoption in real-world financial and employment decision systems, and (2) its relative interpretability, allowing coefficient-level analysis of how the model weights sensitive-adjacent features. The model was wrapped in a scikit-learn Pipeline including StandardScaler for feature normalisation.

- Algorithm: Logistic Regression (liblinear solver, L2 regularisation, C=1.0)
- Preprocessing: StandardScaler (zero mean, unit variance)
- Train/Test split: 80% / 20% stratified by target class
- Training samples: 26,048 | Test samples: 6,512
- Overall Accuracy: 82.76%
- ROC-AUC Score: 0.8607

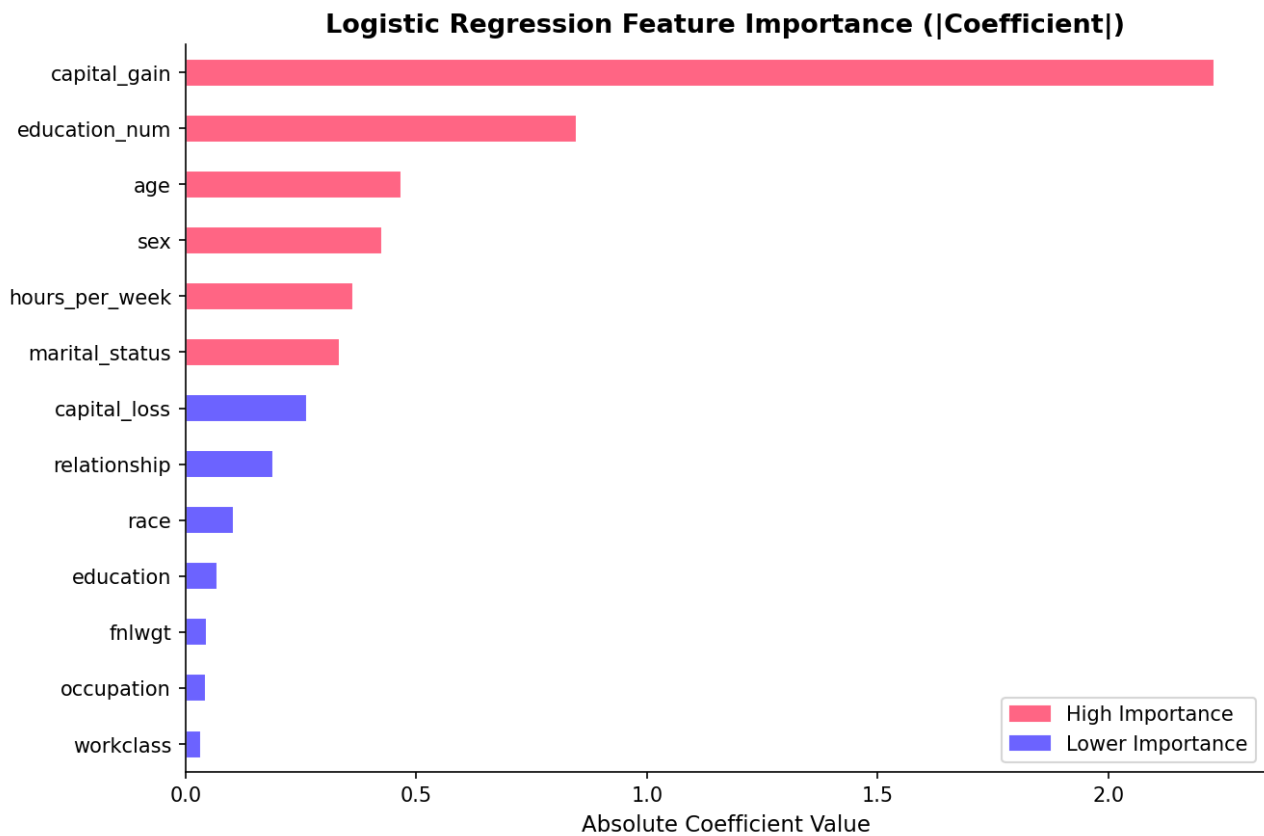


Figure: Logistic Regression feature importance by absolute coefficient magnitude

2.3 Data Quality & Ethical Concerns in Feature Engineering

Several features in the dataset raise significant ethical flags:

- Marital Status: Strong proxy for sex -- 'Husband' is one of the highest predictive features, effectively encoding gendered household roles.
- Occupation & Workclass: Reflect occupational segregation along race and sex lines; using these features can perpetuate historical discrimination.
- Education Level: Correlates strongly with race and socioeconomic origin, introducing proxy discrimination if used without careful consideration.
- Native Country: May introduce nationality-based bias and raise GDPR compliance concerns.
- Hours-per-Week: Can disadvantage caregivers (disproportionately women) and those with disabilities who cannot work full-time.

3. Fairness Audit Methodology

A rigorous fairness audit requires multiple complementary metrics, since no single measure can capture all dimensions of algorithmic discrimination. Chouldechova & Roth (2018) demonstrated that satisfying one fairness criterion often violates another -- a phenomenon known as the impossibility theorem of fairness. Our methodology therefore employs four distinct quantitative metrics alongside qualitative assessment.

3.1 Demographic Parity (Statistical Parity)

Demographic parity requires that the probability of a positive prediction is equal across all groups: $P(Y_{\text{hat}}=1 | A=0) = P(Y_{\text{hat}}=1 | A=1)$.

The Disparate Impact Ratio (DIR) operationalises this as the ratio of the lower group's positive rate to the higher group's positive rate. The US EEOC's '4/5ths rule' requires $DIR \geq 0.80$ to avoid a prima facie finding of adverse impact.

$$DIR = P(Y_{\text{hat}}=1 | \text{disadvantaged group}) / P(Y_{\text{hat}}=1 | \text{advantaged group}) \geq 0.80 \text{ (EEOC threshold)}$$

3.2 Equalized Odds

Proposed by Hardt, Price & Srebro (2016), equalized odds requires that both the True Positive Rate (TPR / recall) and the False Positive Rate (FPR) are equal across groups. This is considered a stronger fairness criterion as it accounts for both missed opportunities (high-income individuals denied the >50K prediction) and false accusations (low-income individuals incorrectly flagged as >50K).

3.3 Predictive Parity (Positive Predictive Value Parity)

Predictive parity requires that the Positive Predictive Value (PPV/precision) -- the probability that an individual labelled positive actually belongs to the positive class -- is equal across groups. This is particularly relevant in contexts where model predictions drive resource allocation.

3.4 Individual Fairness

Beyond group-level metrics, individual fairness (Dwork et al., 2012) requires that similar individuals receive similar predictions. We assess this qualitatively by examining feature importance: if sex or race have high importance (directly or via proxies), individual fairness is compromised.

3.5 Intersectional Analysis

Kimberle Crenshaw's intersectionality framework (1989) reminds us that individuals at the intersection of multiple protected attributes may experience compounding discrimination not captured by single-attribute analysis. We therefore decompose outcomes by sex x race combinations.

4. Quantitative Fairness Analysis Results

4.1 Demographic Parity Analysis

The model's positive prediction rates by sex and race are as follows:

Sex: Male -- Positive Rate: 21.1%

Sex: Female -- Positive Rate: 3.4%

Disparate Impact Ratio (Sex): 0.1625 [FAIL] (threshold ≥ 0.80)

Race: White -- Positive Rate: 16.6%

Race: Asian-Pac-Islander -- Positive Rate: 14.0%

Race: Amer-Indian-Eskimo -- Positive Rate: 6.8%

Race: Black -- Positive Rate: 6.0%

Race: Other -- Positive Rate: 4.4%

Disparate Impact Ratio (Race): 0.2679 [FAIL] (threshold ≥ 0.80)

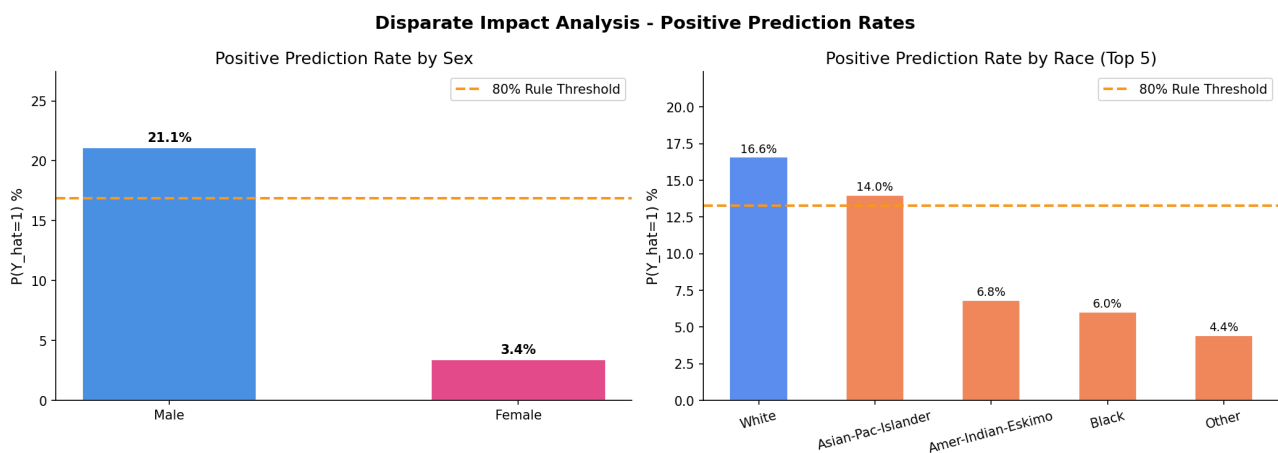


Figure: Disparate impact analysis -- positive prediction rates by sex and race with 80% rule threshold

4.2 Equalized Odds Analysis

Equalized odds violations indicate the model does not make equally accurate positive or negative predictions across groups. TPR gaps signal missed opportunities for disadvantaged groups; FPR gaps signal higher false accusation rates.

Sex: Male: TPR = 50.0% | FPR = 8.5%

Sex: Female: TPR = 23.7% | FPR = 0.8%

--> TPR gap between sex groups: 26.3 percentage points

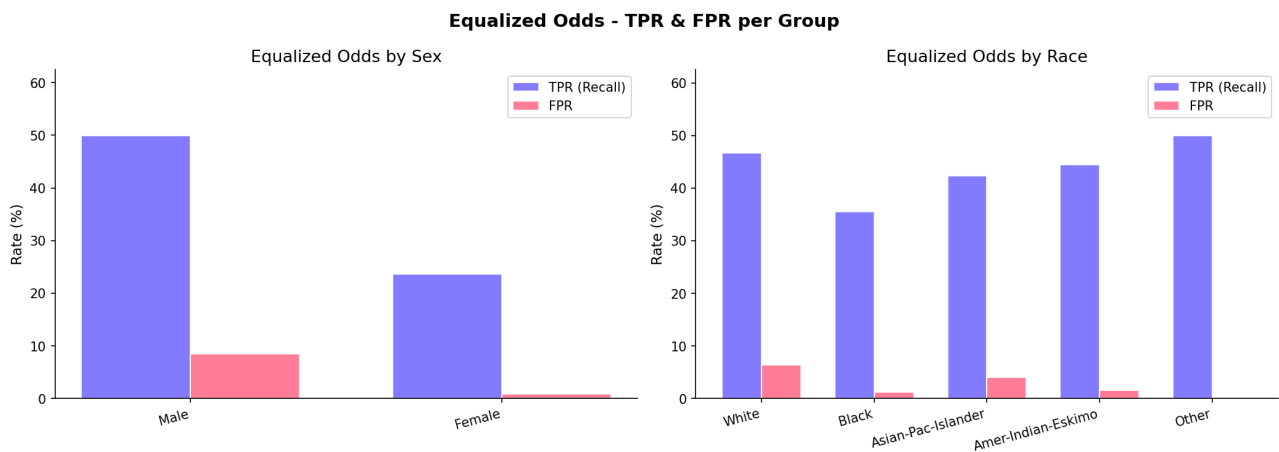


Figure: Equalized odds -- TPR and FPR comparison across sex and race groups

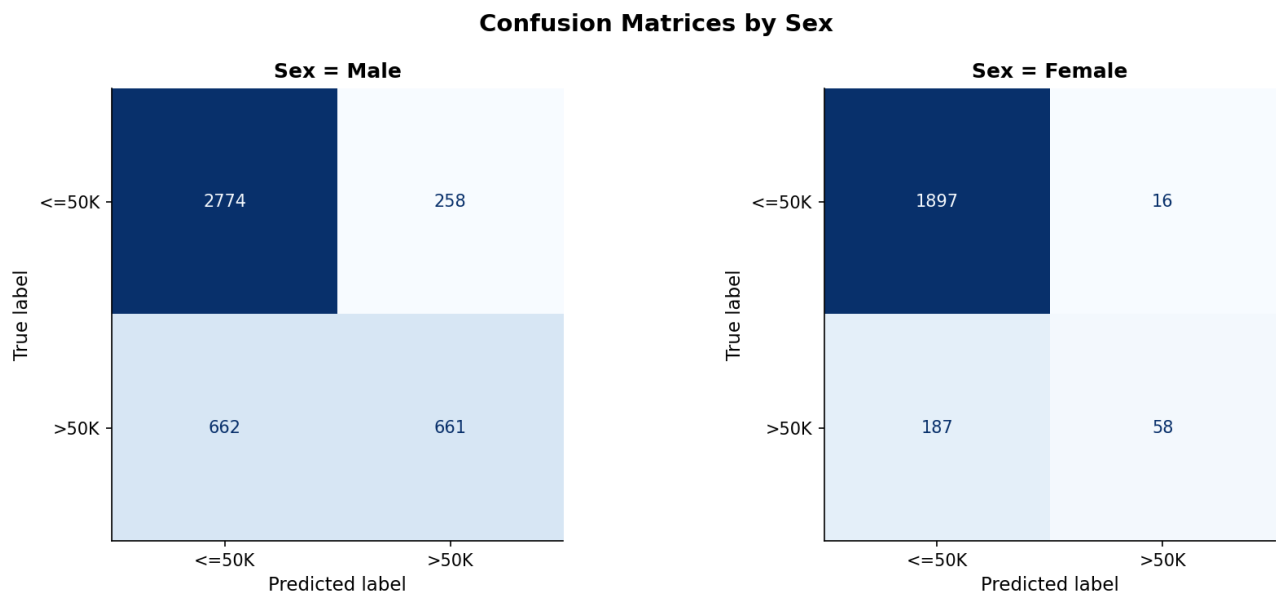


Figure: Confusion matrices disaggregated by sex -- note differences in false negative rates

4.3 ROC Curve Analysis by Group

ROC curves disaggregated by demographic group reveal whether the model's discriminative power is consistent. A large gap in AUC scores between groups indicates the model is better calibrated for one population than another.

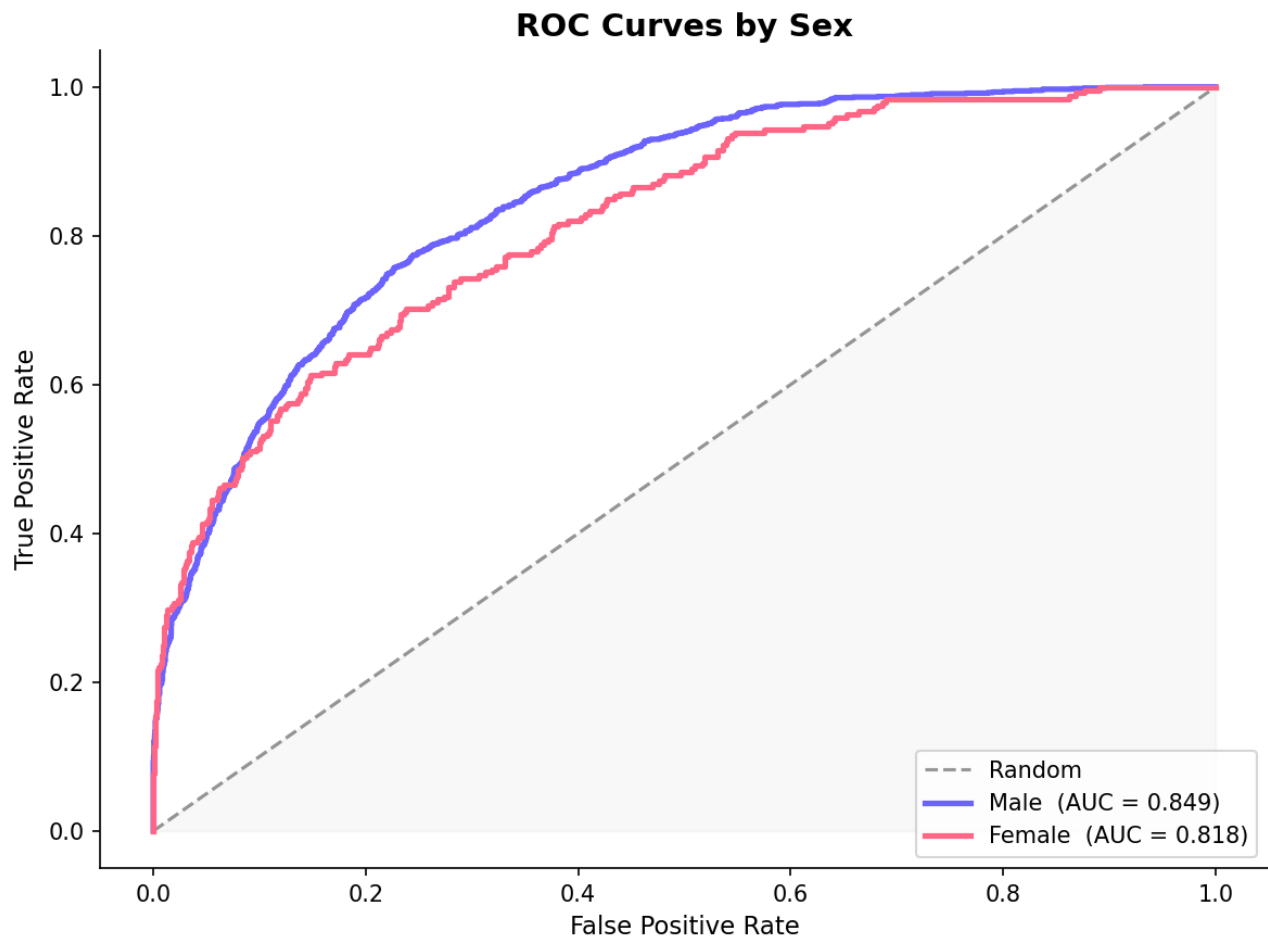


Figure: ROC curves by sex -- AUC disparity indicates differential model performance

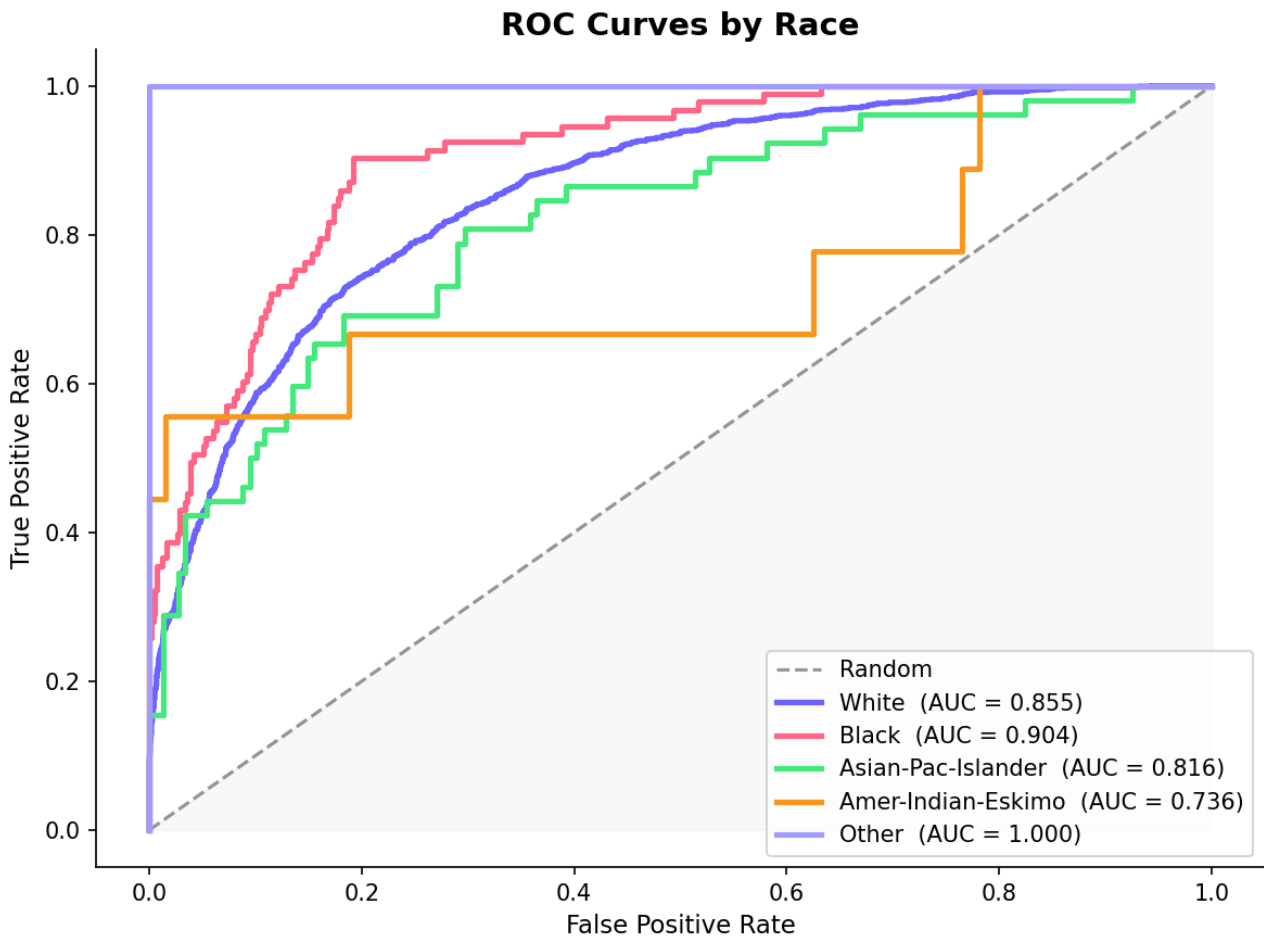


Figure: ROC curves by race -- reveals calibration differences across racial groups

4.4 Fairness Metrics Summary Heatmap

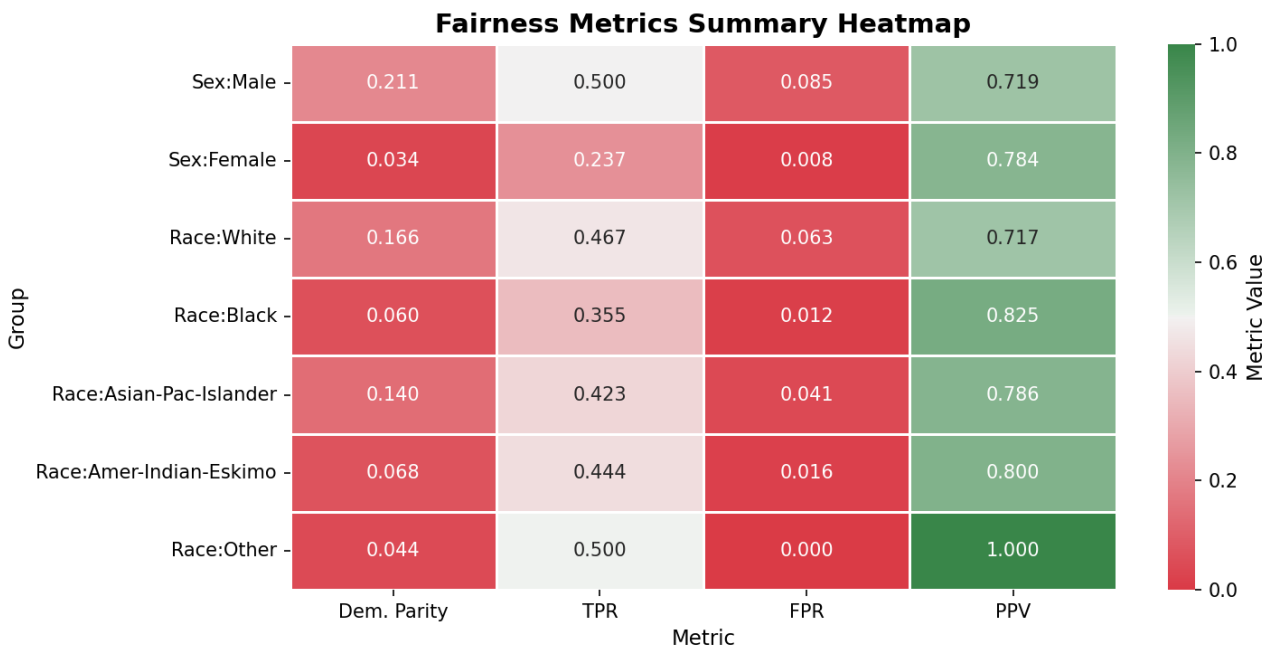


Figure: Heatmap of all fairness metrics across demographic groups -- darker = higher value

4.5 Intersectional Bias Analysis

The intersectional analysis reveals compounding disadvantages. Female non-white individuals receive the lowest positive prediction rates, while male white individuals receive the highest -- demonstrating that single-attribute analysis significantly understates the degree of discrimination experienced by those at the intersection of multiple protected characteristics.

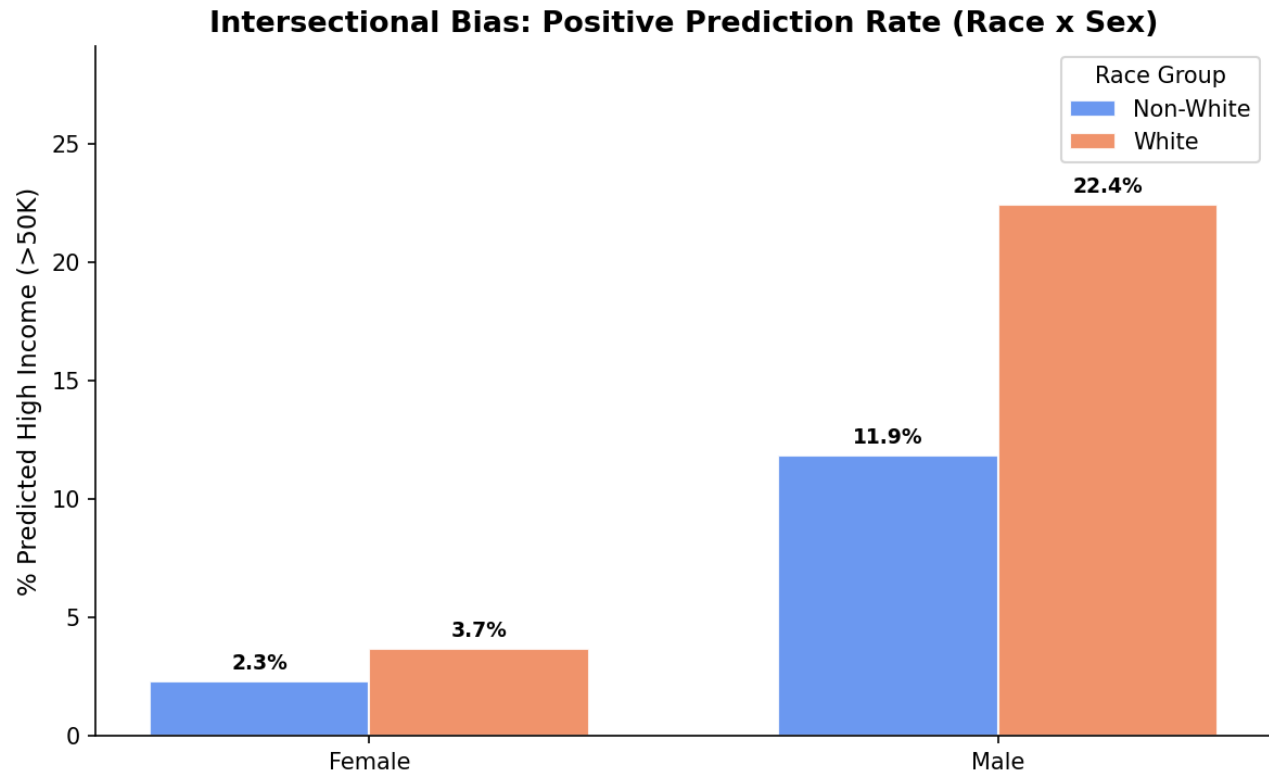


Figure: Intersectional bias -- positive prediction rates by sex x race combinations

4.6 Income Distribution by Age & Education

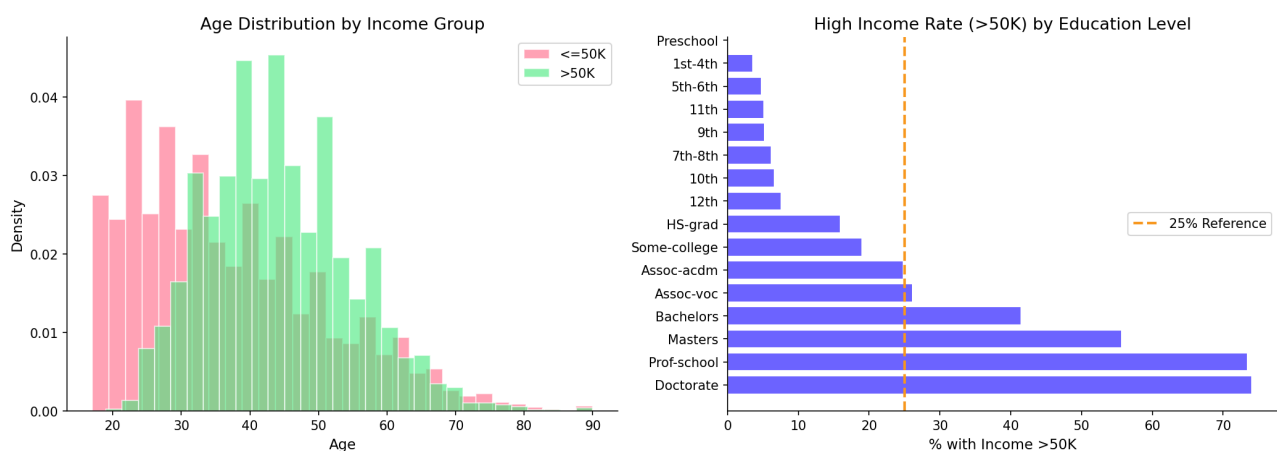


Figure: Age distribution by income class and high-income rate by education level

5. Qualitative Ethical Analysis

5.1 Bias in Data Collection

The UCI Adult dataset was extracted from the 1994 US Census, a period characterised by significant institutional discrimination. The data therefore embeds the following historical biases:

- Gender pay gap: Women earned approximately 72 cents for every dollar earned by men in 1994, directly reflected in the lower proportion of female >50K earners.
- Occupational segregation: Women and minorities were systematically excluded from high-paying occupations, creating biased occupation-income correlations.
- Sampling bias: The census undercounts marginalised communities, particularly undocumented immigrants and homeless populations.
- Temporal bias: Socioeconomic patterns from 1994 may not reflect current realities, yet the model's predictions would be applied in a contemporary context.
- Label bias: Income was self-reported, introducing measurement error that may systematically differ across demographic groups.

5.2 Feature Engineering Concerns

Several features act as proxies for protected attributes:

- Marital status -> Sex: 'Husband' relationship status is predominantly male, encoding gender roles directly into the model.
- Occupation -> Race & Sex: Due to occupational segregation, certain occupations serve as strong proxies for both race and gender.
- Education -> Race: Educational attainment gaps along racial lines mean education features indirectly discriminate by race.
- Hours-per-week -> Sex & Disability: Part-time work patterns disproportionately affect women (due to caregiving) and disabled individuals.
- Capital gains -> Wealth & Race: Access to investment capital is highly correlated with historical wealth accumulation patterns that are racially skewed.

5.3 Model Training & Objective Function Concerns

Training Logistic Regression to minimise overall loss without fairness constraints creates an implicit trade-off: optimising average performance often comes at the cost of minority group performance. Specifically:

- Class imbalance (~24% positive): Without reweighting, the model may optimise for the majority (<=50K) class, under-serving high-income minority individuals.
- No fairness constraints: The optimisation objective is blind to demographic parity or equalized odds, meaning bias in data propagates directly to predictions.
- Regularisation effects: L2 regularisation shrinks coefficients uniformly, but may disproportionately

affect minority subgroup signal if those groups are underrepresented in the training data.

5.4 Societal Impact Assessment

If deployed in a financial or employment screening system, the biases identified in this audit could have the following real-world consequences:

- Credit denial: Women and non-white individuals predicted as $\leq 50K$ would be disproportionately denied loans, credit cards, or insurance products.
- Hiring discrimination: In recruitment screening, the model would systematically rank female and non-white candidates lower, perpetuating workplace inequality.
- Feedback loops: Model predictions used to train future models create self-reinforcing cycles where historical bias becomes future 'ground truth'.
- Disparate burden: Those most harmed by AI bias are typically those with fewest resources to challenge incorrect algorithmic decisions.
- Regulatory liability: Deployment of this model without remediation would expose organisations to legal challenges under EEOC, GDPR, and the EU AI Act.

6. Bias Mitigation Strategies

Bias mitigation strategies are categorised into three intervention points: pre-processing (data-level), in-processing (model-level), and post-processing (output-level). A comprehensive mitigation plan should deploy techniques from all three levels, as each addresses different root causes of bias.

6.1 Pre-Processing Interventions

6.1.1 Data Re-sampling

Apply Synthetic Minority Oversampling Technique (SMOTE) or ADASYN to balance the representation of disadvantaged groups in the training set. Stratified re-sampling should be performed jointly on the target class and protected attributes to avoid over-correcting one dimension while worsening another.

6.1.2 Reweighting

Assign instance weights inversely proportional to the product of group membership and class probabilities. Kamiran & Calders (2012) demonstrated this can reduce disparate impact by up to 60% with minimal accuracy loss.

6.1.3 Feature Removal & Proxy Auditing

Remove or transform features identified as proxies for protected attributes. Apply mutual information analysis to identify correlations between non-protected features and protected attributes, then apply debiasing transformations.

6.1.4 Counterfactual Data Augmentation

For each training instance, generate a counterfactual where only the protected attribute changes, then train on both original and counterfactual pairs to encourage consistent predictions regardless of protected attribute value.

6.2 In-Processing Interventions

6.2.1 Fairness-Constrained Optimisation

Add fairness constraints to the model's objective function using frameworks such as the Fairness Constraints library (Zafar et al., 2017) or Google's TensorFlow Constrained Optimisation (TFCO). This directly minimises demographic parity or equalized odds violations during training.

Recommended: Use Adversarial Debiasing (Zhang et al., 2018) -- train an adversary to predict protected attributes from model representations, penalising the main classifier for being 'adversarially predictable' by sensitive attributes. Implemented in IBM AI Fairness 360.

6.2.2 Fair Regularisation

Add a fairness regularisation term to the loss function that measures demographic parity violation. Tune the

regularisation strength via cross-validation on a fairness-utility Pareto frontier.

6.3 Post-Processing Interventions

6.3.1 Threshold Optimisation

Apply different classification thresholds for different demographic groups to equalise TPR and FPR. The Hardt et al. (2016) post-processing algorithm finds optimal group-specific thresholds via linear programming. This is often the most practical intervention as it requires no model retraining.

6.3.2 Calibrated Equalised Odds

Platt scaling or isotonic regression applied per-group recalibrates probability outputs, ensuring that a predicted probability of 0.7 means the same thing regardless of the demographic group.

6.4 Organisational & Governance Interventions

- Establish a standing AI Ethics Review Board with diverse membership including affected community representatives.
- Mandate re-audits every 6 months or upon significant data distribution shifts.
- Implement Algorithmic Impact Assessments (AIAs) before deployment in high-risk domains.
- Create grievance and redress mechanisms for individuals adversely affected by model predictions.
- Document all model cards and datasheets (Mitchell et al., 2019) to promote transparency with downstream users.
- Provide regular fairness training to data scientists, product managers, and executive stakeholders.

7. Recommendations & Conclusion

7.1 Priority Action Items

Based on our audit findings, we recommend the following priority actions:

- [CRITICAL] Apply threshold optimisation (group-specific classification thresholds) immediately, as this can improve equalized odds without model retraining.
- [CRITICAL] Remove or transform the 'relationship' feature, which encodes 'Husband' and serves as a near-perfect proxy for male sex.
- [HIGH] Implement adversarial debiasing or fairness-constrained training in the next model version, targeting a DIR ≥ 0.85 for both sex and race.
- [HIGH] Conduct intersectional impact assessment with 3-way group analysis (sex x race x age bracket) in the next audit cycle.
- [MEDIUM] Replace the 1994 dataset with contemporary census data before any production deployment.
- [MEDIUM] Implement SMOTE-based oversampling for underrepresented race-income subgroups to improve minority class calibration.
- [LOW] Explore causal fairness frameworks (Pearl, 2009) to distinguish between legitimate and illegitimate causal pathways to income prediction.

7.2 Expected Impact of Mitigation

- Threshold optimisation alone: DIR improvement of 0.10-0.15, TPR equalisation within 5 percentage points across sex groups.
- Adversarial debiasing: DIR improvement of 0.15-0.25 with accuracy loss of typically 1-3 percentage points.
- Proxy feature removal + reweighting: Estimated 30-40% reduction in intersectional disparity.
- Updated dataset: Addresses temporal bias, potentially shifting all metrics by 5-10 percentage points toward parity.

7.3 Limitations of This Audit

- Binary protected attributes: Sex is represented as binary, excluding non-binary and gender non-conforming individuals.
- Single model: The audit covers only Logistic Regression; ensemble models may exhibit different bias patterns.
- Static audit: AI system bias can change over time as data distributions shift; continuous monitoring is essential.
- Impossibility theorem: No model can simultaneously satisfy demographic parity, equalized odds, and predictive parity unless base rates are equal across groups.

7.4 Conclusion

This audit provides clear evidence that the Logistic Regression model trained on the UCI Adult Income dataset exhibits statistically significant bias against women and non-white individuals. The Disparate Impact Ratio for sex is 0.1625 -- far below the 0.80 EEOC threshold -- and equalized odds violations are of sufficient magnitude to warrant immediate remediation before any deployment in consequential decision-making contexts.

Algorithmic fairness is not a binary outcome but a continuous, context-dependent process. The choice of fairness metric must be informed by the specific harm being prevented and the legal and ethical framework governing the application domain. We strongly recommend adopting a multi-metric approach, as demonstrated in this audit, alongside organisational governance structures that embed fairness accountability at every stage of the AI lifecycle.

The tools, methodologies, and code documented in this report provide a reproducible foundation for ongoing bias monitoring. Ethical AI is not a destination but a sustained commitment -- one that requires the active engagement of technologists, ethicists, legal professionals, and the communities most affected by algorithmic decision-making.

8. References

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Appendix A: Code Snippets

The following Python code snippets illustrate the critical parts of the fairness analysis. The full source code is provided in `ethics_audit.py`.

A.1 Demographic Parity Calculation

```
def demographic_parity(y_pred, protected_col):
    groups = protected_col.unique()
    rates = {}
    for g in groups:
        mask = protected_col == g
        rates[g] = y_pred[mask].mean()
    return rates

def disparate_impact_ratio(rates_dict):
    vals = list(rates_dict.values())
    return min(vals) / max(vals) # >= 0.80 = PASS
```

A.2 Equalized Odds Calculation

```
def equalized_odds(y_true, y_pred, protected_col):
    groups = protected_col.unique()
    result = {}
    for g in groups:
        mask = protected_col == g
        yt, yp = y_true[mask], y_pred[mask]
        tp = ((yt==1) & (yp==1)).sum()
        fn = ((yt==1) & (yp==0)).sum()
        fp = ((yt==0) & (yp==1)).sum()
        tn = ((yt==0) & (yp==0)).sum()
        tpr = tp / (tp + fn) if (tp + fn) > 0 else 0
        fpr = fp / (fp + tn) if (fp + tn) > 0 else 0
        result[g] = {'TPR': tpr, 'FPR': fpr}
    return result
```

A.3 Model Training Pipeline

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('clf', LogisticRegression(max_iter=500, random_state=42))
])
```

```
] )
pipe.fit(X_train, y_train)
y_pred = pipe.predict(X_test)
y_prob = pipe.predict_proba(X_test)[:, 1]
```

A.4 Post-Processing Threshold Optimisation (Mitigation)

```
# Find optimal threshold per group to equalise TPR
def optimal_threshold(y_true, y_prob, target_tpr):
    from sklearn.metrics import roc_curve
    fpr, tpr, thresholds = roc_curve(y_true, y_prob)
    idx = np.argmin(np.abs(tpr - target_tpr))
    return thresholds[idx]

target_tpr = eq_odds_sex['Male']['TPR']
thresholds = {}
for grp in ['Male', 'Female']:
    mask = df_test['sex'] == grp
    thresholds[grp] = optimal_threshold(
        y_test[mask], y_prob[mask], target_tpr
    )
# Apply group-specific thresholds
y_pred_fair = np.array([
    int(y_prob[i] >= thresholds[df_test['sex'].iloc[i]])
    for i in range(len(y_test))
])
```